ECET 250E-Linear Circuit 1
Quiz 1-Chapter 1&2- Basic Concepts & Laws
Solution

Name: _____

Total Marks: _____ / 30	**Percentage:** _____ %

1. The dependent source in figure below is:

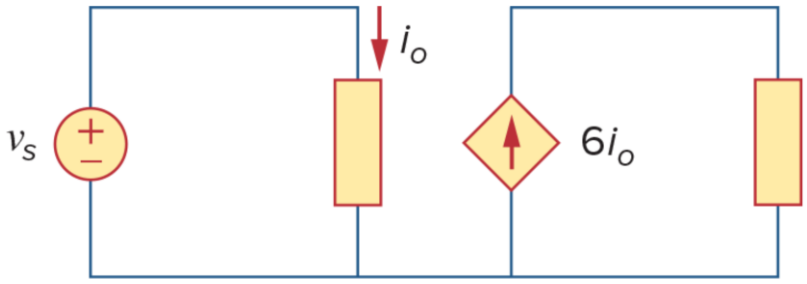
(a)voltage-controlled current source

(b)voltage-controlled voltage source

(c)current-controlled voltage source

(d)current-controlled current source

Answer: *d*



2. Determine the current flowing through an element if the charge flow is given by $\boldsymbol{q(t) = (9e^{-3t} - 2e^{-6t})\ nC}$

[Hint: i = dq/dt]

$i = dq/dt = d/dt\ (9e^{-3t} - 2e^{-6t}) = (-27e^{-3t}) + (12e^{-6t}) =$

$= (-27e^{-3t} + 12e^{-6t})\ nA$ *Marks (2): 1 for each correct derivative term.*

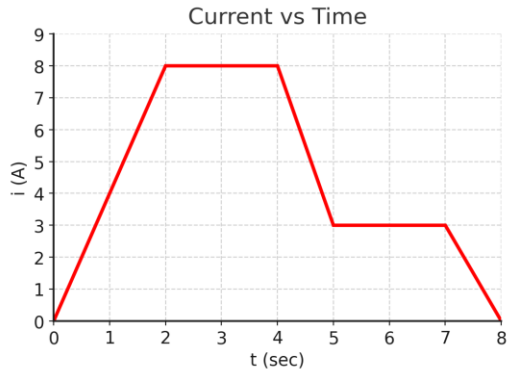
3. Determine the **total charge** transferred over the time interval of **0 ≤ t ≤ 6s** when $\boldsymbol{i(t) = \frac{1}{3}t}$ **A**.

Solution:

$$q = \int idt = \int_0^6 \frac{1}{3} t dt = \frac{1}{3}\left(\frac{t^2}{2}\right)\Big|_0^6 = \frac{1}{3}\left(\frac{6^2}{2} - \frac{0^2}{2}\right) = \frac{1}{3} * \frac{36}{2} = \frac{1}{3} * 18 = 6 \text{ C}$$

Marks (2): 1 for the correct integration, 1 for applying the limits and the final answer (6 C).

4. The current flowing past a point in a device is shown in Fig. below. Calculate the total charge through the point



Solution:

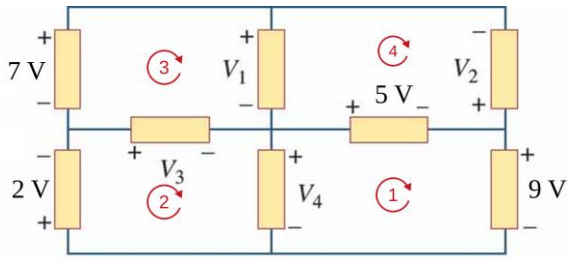
$$q = \int \text{idt} = A1 + A2 + A3 + A4 + A5$$

$$= \left(\frac{1}{2} \times 2 \times 8\right) + (2 \times 8) + \left(\frac{1}{2} \times 1 \times 5\right) + (3 \times 3)$$

$$+ \left(\frac{1}{2} \times 1 \times 3\right) = 8 + 16 + 2.5 + 9 + 1.5 = \underline{37\ C}$$

Marks (4): 3 for the areas A1 to A5, 1 for the total charge with units (37 C).

5. Given the circuit in Fig. below, use KVL to find the branch voltages V_1 to V_4.



For mesh 1,

$$-V_4 + 5 + 9 = 0$$

$$V_4 = 14 \text{ V}$$

For mesh 2,

$$+2 + V_3 + V_4 = 0$$

$$V_3 = -2 - 14 = -16 \text{ V}$$

For mesh 3,

$$-7 + V_1 - V_3 = 0$$

$$V_1 = V_3 + 7 = -16 + 7 = -9 \text{ V}$$

For mesh 4,

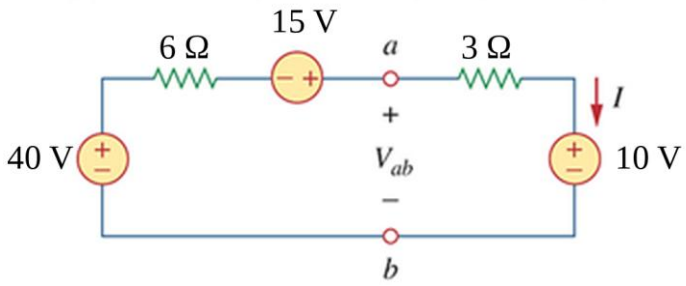
$$-V_1 - V_2 - 5 = 0$$

$$V_2 = -V_1 - 5 = -(-9) - 5 = 9 - 5 = 4 \text{ V}$$

Thus, $V_1 = -9$ V, $V_2 = 4$ V, $V_3 = -16$ V, $V_4 = 14$ V

Marks (4): 1 for each branch voltage V_1 to V_4 with its KVL equation.

6. Find I and V_{ab} in the circuit of Fig. Below.



Applying KVL,

-40 -15 +10 + I (6+3) = 0

9I = 45 I = **5A**

$-V_{ab} + 3I + 10 = 0$ $-V_{ab} + (3*5) + 10 = 0$ $V_{ab} =$ **25V**

Marks (4): 1 for the KVL loop equation, 1 for I, 1 for the V_{ab} equation, 1 for V_{ab}.

7. The total power dissipated in a parallel circuit is equal to the

a. power in the largest resistor.

b. power in the smallest resistor.

c. average of the power in all resistors.

d. sum of the power in all resistors.

answer d

8. KCL is based on the law of conservation of

a. energy.

b. charge.

c. voltage.

d. power.

answer b

9. Which of the following ideal circuit elements dissipates energy

a. inductor.

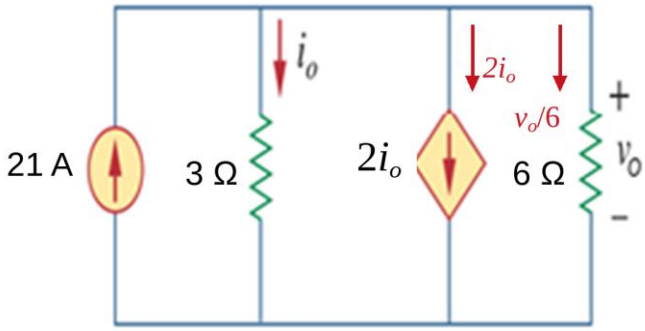
b. capacitor.

c. resistor.

d. none of the above.

answer c

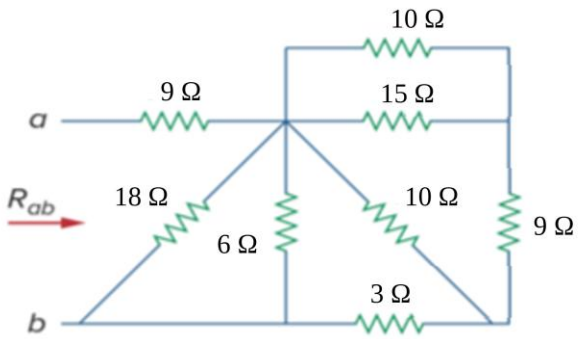
10. Find V_o, I_o in the circuit below:



Applying KCL, $21 - i_o - [2i_o] - [v_o/6] = 0$, but $i_o = v_o/3$
Which leads to: $21 = (v_o/3) + (2v_o/3) + (v_o/6)$ thus, $v_o =$ **18 V** and $i_o =$ **6 A**

Marks (4): 1 for the KCL equation, 1 for the control relation $i_o = v_o/3$, 1 for v_o, 1 for i_o.

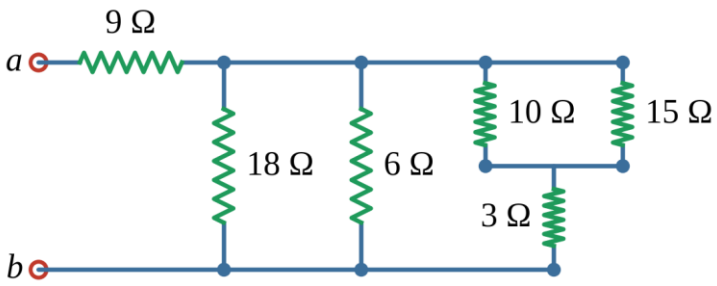
11. Find $\mathbf{R_{ab}}$ (Resistor Equal) for the circuit below.



Solution:

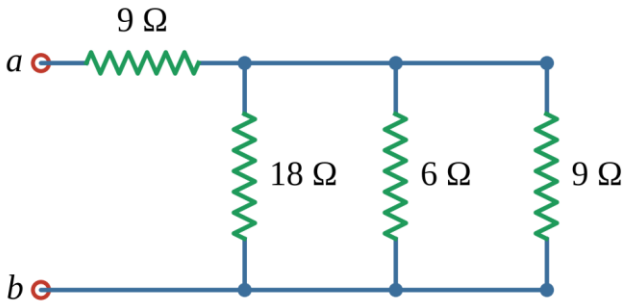
Step 1: The top 10 Ω and the middle 15 Ω resistors are in parallel: $10 \parallel 15 = (10\times15)/(10+15) = 150/25 = 6\ \Omega$

Step 2: That 6 Ω is in series with the 9 Ω on the right: $6 + 9 = 15\ \Omega$



Step 3: This 15 Ω is in parallel with the 10 Ω (diagonal) resistor: 10 || 15 = 150/25 = 6 Ω

Step 4: That 6 Ω is in series with the bottom 3 Ω: 6 + 3 = 9 Ω



Step 5: The 18 Ω, 6 Ω and 9 Ω branches are all connected between the same two nodes, so they are in parallel: $1/R = 1/18 + 1/6 + 1/9 = (1 + 3 + 2)/18 = 6/18$ → $R = 3\ \Omega$

Step 6: This 3 Ω is in series with the 9 Ω connected to terminal a: $R_{ab} = 9 + 3 =$ **12 Ω**

Marks (6): 1 for each step (Steps 1 to 6).